Is masked image modeling effective for remote sensing foundation models?

Yes, **masked image modeling works** for **remote sensing foundation models**, though performance depends on task and design.

Is masked image modeling effective for remote sensing foundation models? N = 17

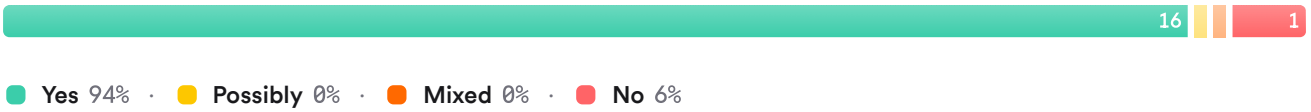


FIGURE 1 Research consensus on masked image modeling effectiveness

The literature is **largely supportive** rather than mixed. Across surveys, foundation-model papers, and task-specific studies, masked image modeling consistently improves remote-sensing representation learning, transfer, and scalability, but standard MAE-style masking often underperforms on small objects, dense scenes, and low-label dense prediction unless adapted to remote-sensing structure (Sun et al., 2023; Lu et al., 2025; Lu et al., 2024; Choudhury et al., 2025; Liu et al., 2026).

Overall Answer

Yes. MIM is repeatedly described as a **core or essential** pretraining method for remote-sensing foundation models (Liu et al., 2023; Wang et al., 2024; Liu et al., 2026).

The strongest direct evidence comes from in-domain foundation-model studies that report state-of-the-art or broad downstream gains after MIM pretraining (Sun et al., 2023; Dong et al., 2024; Wang et al., 2024; Li et al., 2024).

Surveys reach the same conclusion, stating that self-supervised pretraining methods such as MAE and MIM markedly improve the performance and robustness of remote-sensing foundation models (Lu et al., 2025; Lu et al., 2024).

Where It Helps

MIM is effective because remote sensing has abundant **unlabeled imagery**, making generative self-supervision more practical than purely supervised pretraining (Sun et al., 2023; Lu et al., 2024; Choudhury et al., 2025).

It also fits remote sensing because reconstruction objectives encourage **object-level and pixel-level features**, which are important for classification, segmentation, detection, and change-related tasks (Zhang et al., 2023; Choudhury et al., 2025; Wang et al., 2024).

Several studies report concrete downstream gains from MIM variants: SatMAE and Scale-MAE improved land-cover classification and segmentation in survey syntheses, FG-MAE improved SAR performance by up to 5% over standard MAE, and SelectiveMAE improved efficiency by more than 2× while also improving classification, detection, and segmentation (Lu et al., 2024; Wang et al., 2023; Wang et al., 2024).

Remote-sensing-specific MIM designs also help broaden applicability across modalities and use cases, including multispectral, hyperspectral, SAR, cross-sensor retrieval, and season-aware multimodal modeling (Lin et al., 2023; Hackstein et al., 2024; Li et al., 2024; Zhang et al., 2026).

Representative Evidence

Model or Paper	MIM-Related Contribution	Reported Outcome
RingMo	RS foundation model built with MIM on **2 million** images	**SOTA on 8 datasets** across 4 tasks (Sun et al., 2023)
GeoSense SMLFR	ConvNet MIM with sparse modeling and low-frequency targets	**SOTA across 5 datasets** and 3 tasks (Dong et al., 2024)
SelectiveMAE	Efficient MIM on **OpticalRS-13M**	**>2× faster** with better cls/det/seg (Wang et al., 2024)
FG-MAE	Feature-guided reconstruction for RS	**Up to 5%** better than MAE on EuroSAT-SAR (Wang et al., 2023)
OCMIM	Object-centric masking for detection	Better pretraining for **6 detection benchmarks** (Zhang et al., 2023)

FIGURE 2 Representative masked image modeling results in remote sensing

Main Limits

The evidence is not that vanilla MIM is universally best. Multiple papers argue that **random high-ratio masking** can erase small objects and fine local structure in dense remote-sensing scenes, which hurts convergence and dense prediction performance (Zhang et al., 2023; Liu et al., 2026).

Several studies therefore modify standard MAE with context guidance, object-centric masking, spectral masking, edge priors, or semantic token selection to better match remote-sensing data (Zhang et al., 2023; Lin et al., 2023; Liu et al., 2026; Wang et al., 2024).

There are also task-dependent caveats. One recent study notes that MAE- and ViT-based remote-sensing foundation models can be less effective than a basic UNet on some image-to-image tasks, and low-label segmentation work argues that standard MIM objectives are misaligned with fine-grained dense prediction unless adapted (Duc & Fukui, 2025; Liu et al., 2026).

Masked image modeling is effective for remote sensing foundation models overall, but its **best results come from remote-sensing-specific variants** rather than off-the-shelf MAE alone. The consensus is strong on usefulness, and the main open question is how to design masking and reconstruction targets for dense, multisensor, and low-label settings (Sun et al., 2023; Choudhury et al., 2025; Hackel et al., 2025; Li et al., 2024).

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at https://consensus.app. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

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